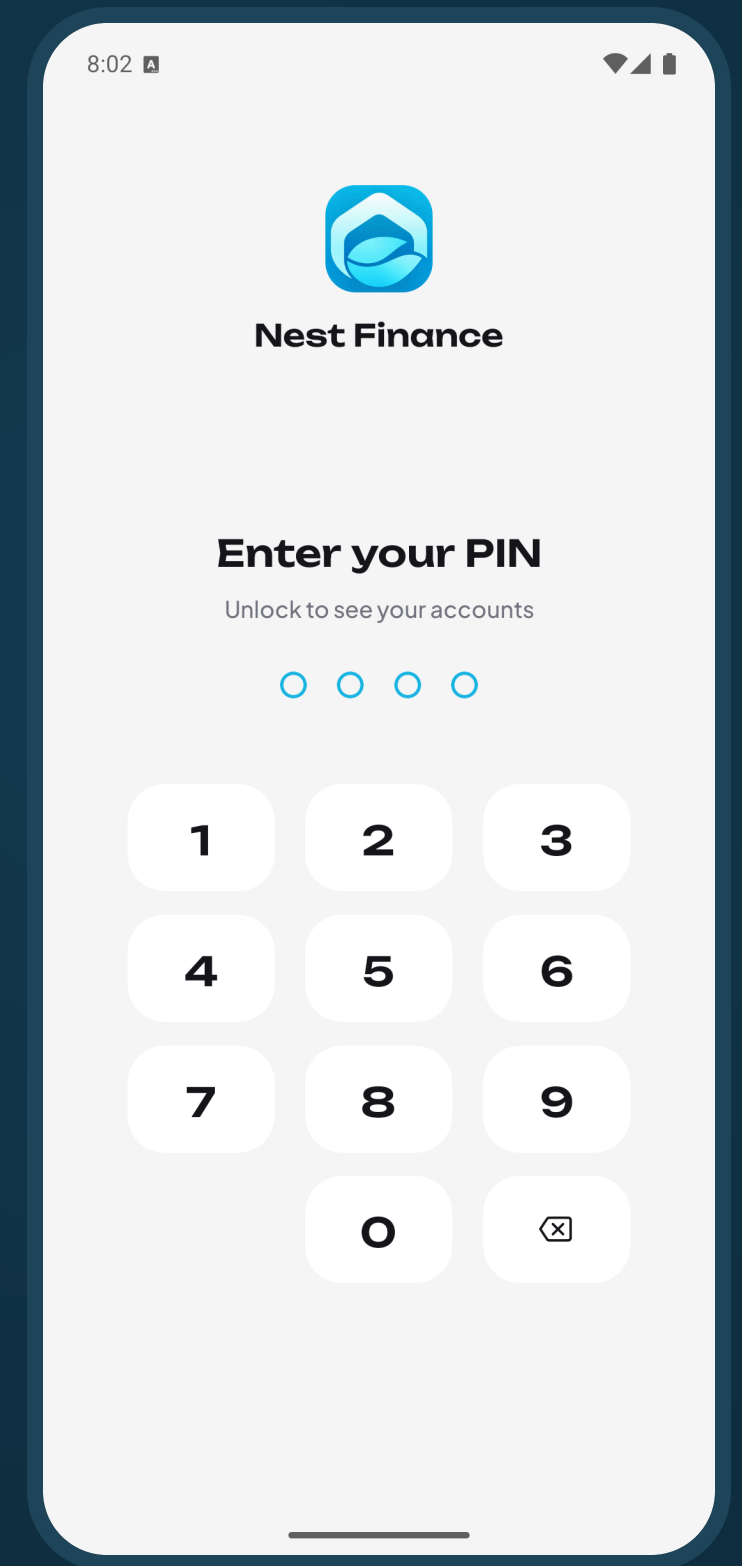




Nest Finance

Savings that can't be rushed.

A time-locked savings vault with a backup PIN and guardians, built for Solana Seeker.



THE PROBLEM

A Seeker in your hand says you hold crypto.

Forced to unlock

Someone next to you makes you open your wallet. Seed phrases and hardware keys don't help when they have you and your phone.

One bad signature

A drainer needs a single approval to empty a wallet.

No time to react

Crypto moves instantly. By the time anyone knows, the money is gone.

THE IDEA

Even with your real key, nobody can rush your savings.

Time-locked vault

Every withdrawal waits a delay you choose, enforced on-chain.

Backup PIN

Opens an ordinary wallet while your savings freeze in the background.

Guardians

People you trust can stop a withdrawal inside the window.

HOW IT WORKS

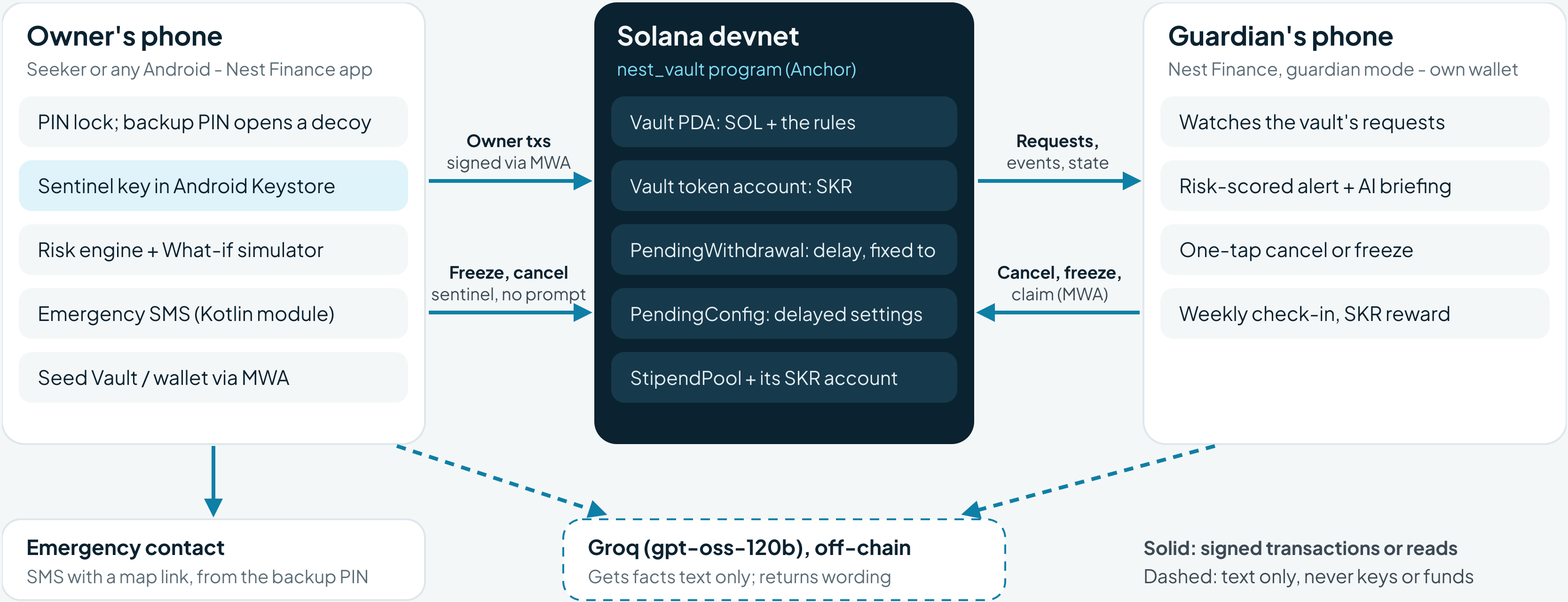
The program enforces the rules, not the app.

- Every withdrawal waits your delay, 24 hours to 7 days.
- Instant moves go only to your own safe addresses.
- Settings changes wait the same delay.
- A freeze voids every pending request at once.
- Early release needs you and a guardian together.

Key	Lives in	Can
Owner	Seed Vault	Deposit, request, cancel
Guardian	Their phone	Cancel, freeze, check in
Sentinel	This phone's keystore	Only freeze and cancel

ARCHITECTURE

Two phones, one program. The chain holds the money and the rules.



ON-CHAIN PROGRAM

5 account types, 17 instructions, each with fixed signers.

Accounts (program-derived)

Vault	["vault", owner]: SOL savings, roles, delay, freeze, epoch
Vault token account	Savings in SKR or another SPL token; only the vault PDA signs
PendingWithdrawal	["withdrawal", vault, id]: amount, fixed destination, unlock time, epoch
PendingConfig	["config", vault]: proposed settings, apply time, epoch
StipendPool	["stipend", vault]: SKR mint, weekly rate, last claim per guardian

Instructions, by who must sign

Owner	init_vault, request_withdrawal, instant_withdraw (safe list only), propose_config, setup_stipend
Owner + guardian	expedite_withdrawal, lift_lockdown
Owner, guardian or sentinel	cancel_withdrawal, cancel_config, lockdown
Guardian	claim_stipend, guardian_heartbeat
Anyone	deposit_sol, deposit_token, fund_stipend; execute_withdrawal and apply_config only after the delay

Only three instructions move savings out: execute (after the delay, to the stored destination), expedite (owner and guardian together) and instant (safe list only). A freeze bumps the epoch, voiding every pending request.

SECURITY MODEL

The delay cannot be skipped, even with the owner's real key.

- The program sets the unlock time, not the caller.
- Payout goes only to the destination stored at request time.
- Instant payouts only reach safe addresses already on the list.
- Adding a safe address, shortening the delay or swapping a guardian waits the current delay, and any role can cancel it.
- A freeze voids every pending request; lifting it early needs owner and guardian together.

If this is compromised	They can	They cannot
The unlocked phone	Ask the wallet to sign a request; freeze	Skip the delay, or reach the owner key
The owner's key	Request, propose, pay out to existing safe addresses	Pay a new address before the delay; change settings faster
One guardian	Cancel and freeze	Move, redirect or release any savings
The sentinel key	Freeze and cancel	Move funds or change settings
Upgrade authority	Deploy new rules	Nothing, once made immutable (before mainnet)

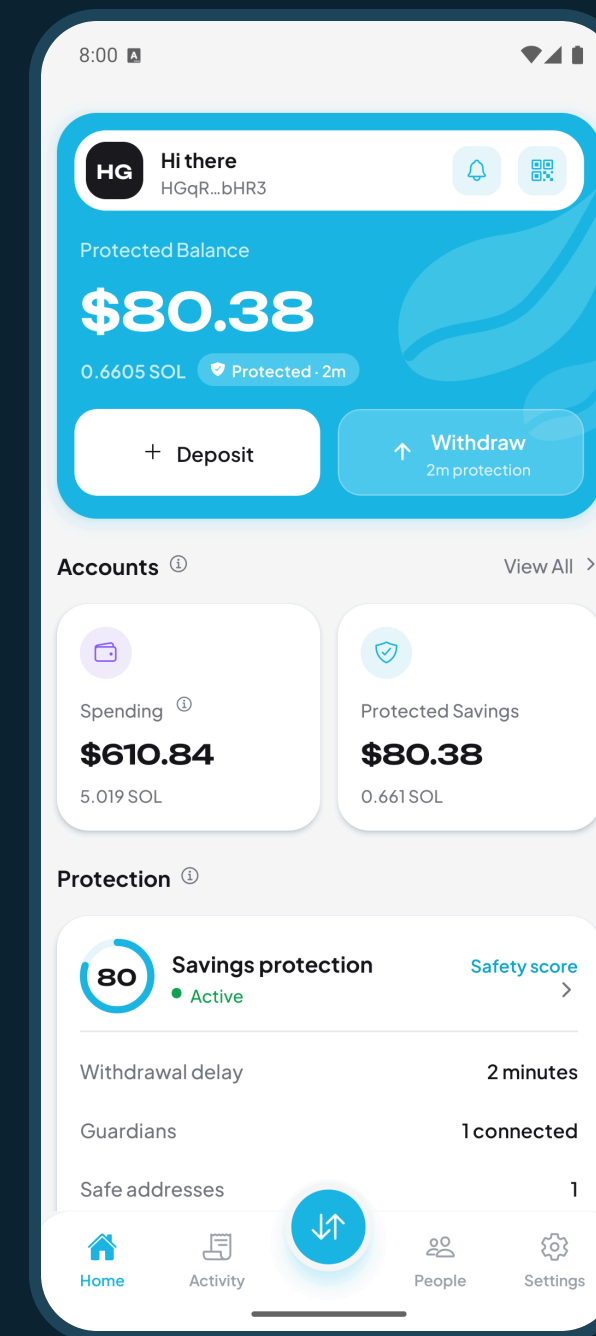
Known gap: a hostile guardian can keep re-freezing. Savings are held up, never taken; a per-role freeze cooldown is planned. Each rule has a test, and 22 of 22 pass on the deployed binary.

THE BACKUP PIN

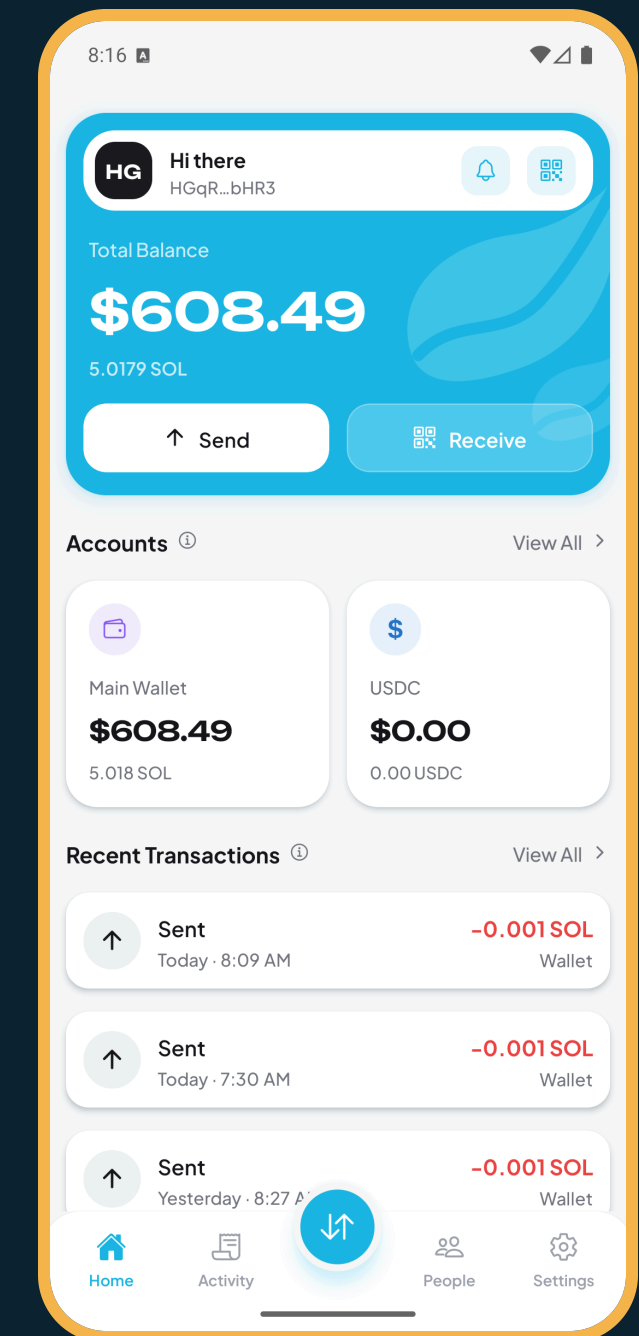
Two PINs. One opens an ordinary wallet.

Both look and behave the same. The backup PIN shows only your spending wallet, real balance and real history, while in the background:

- Savings freeze on-chain, signed by this phone's own key. No wallet prompt.
- Your emergency contact gets a text with your location.
- Any pending withdrawal is void for good.



Your PIN



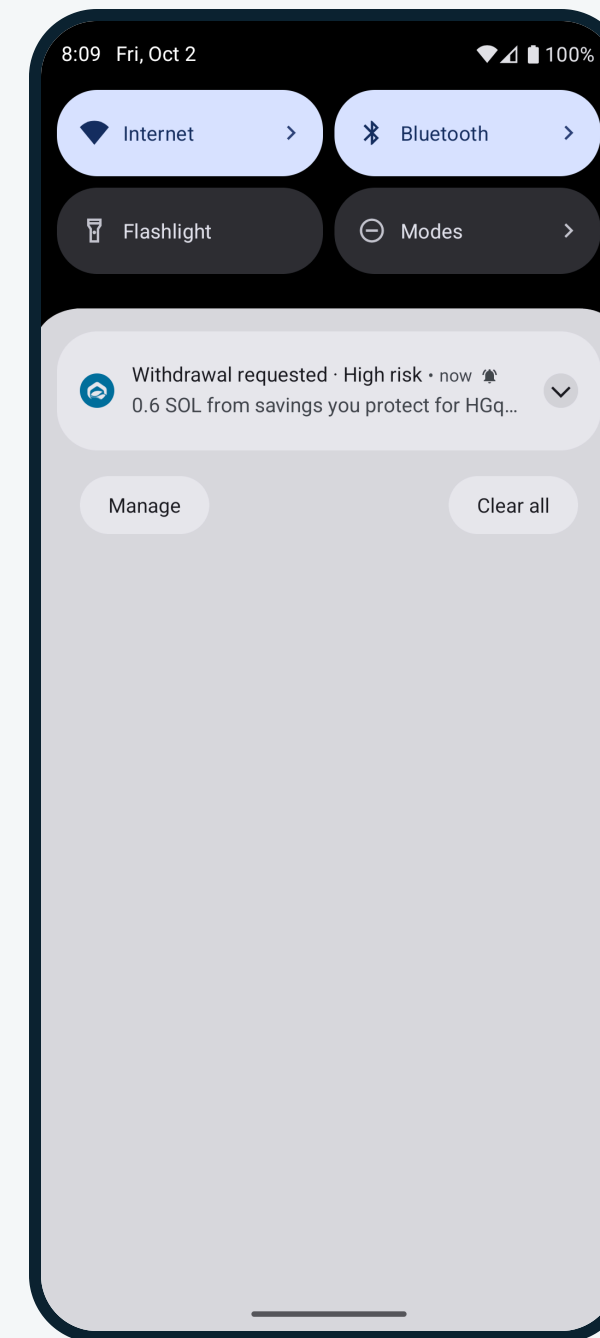
Backup PIN

GUARDIANS

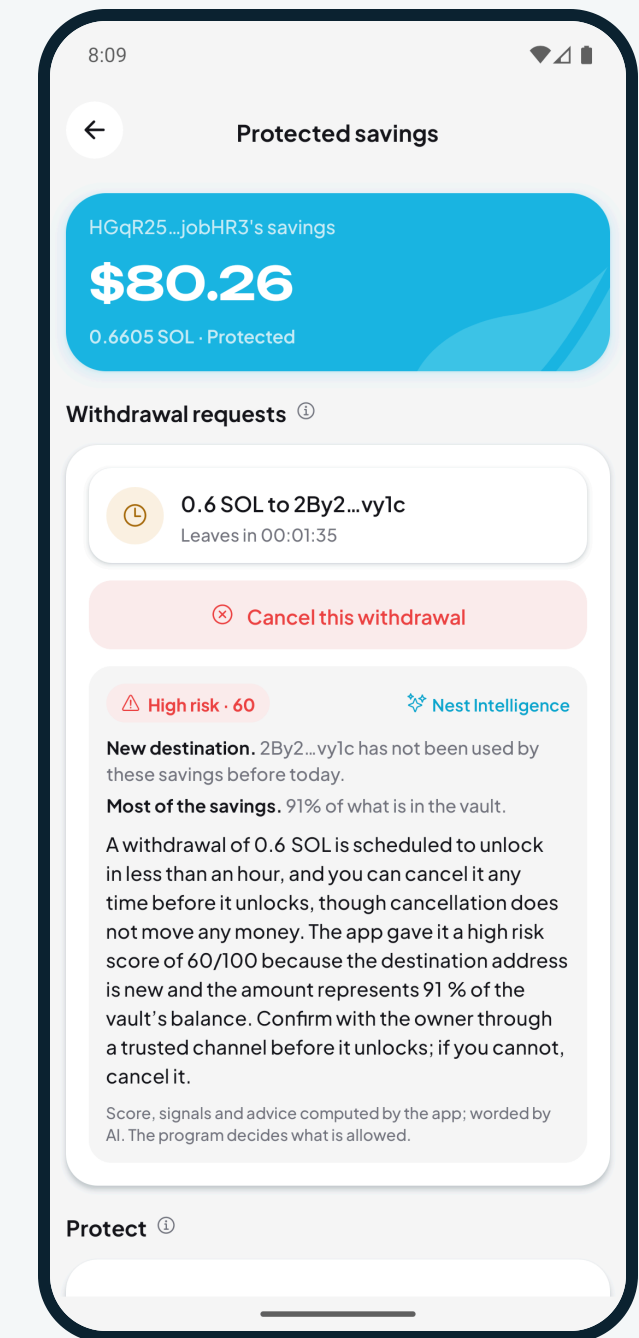
Someone you trust can stop it in time.

- Same app, their own wallet. Found automatically, no invite handshake.
- Alerted when a withdrawal is requested, or the backup PIN is used.
- Can cancel, freeze and check in. Can never take money out.

In testing, the alert arrived about 5 seconds after an outside request.



The alert



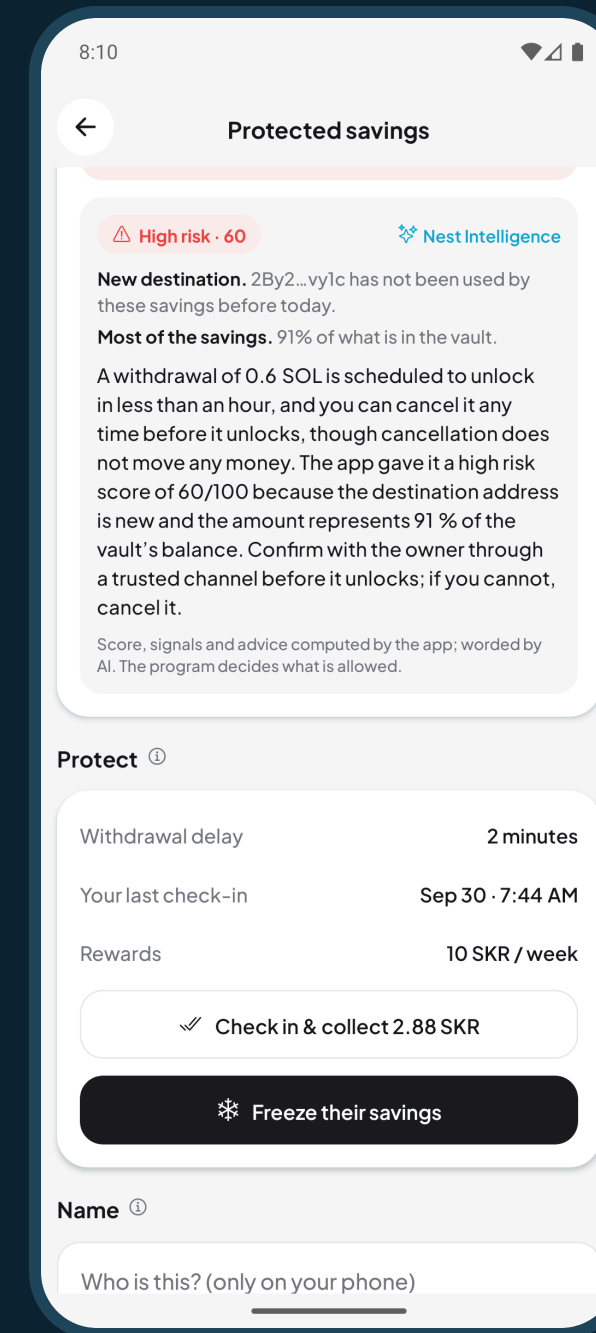
One tap to cancel

NEST INTELLIGENCE

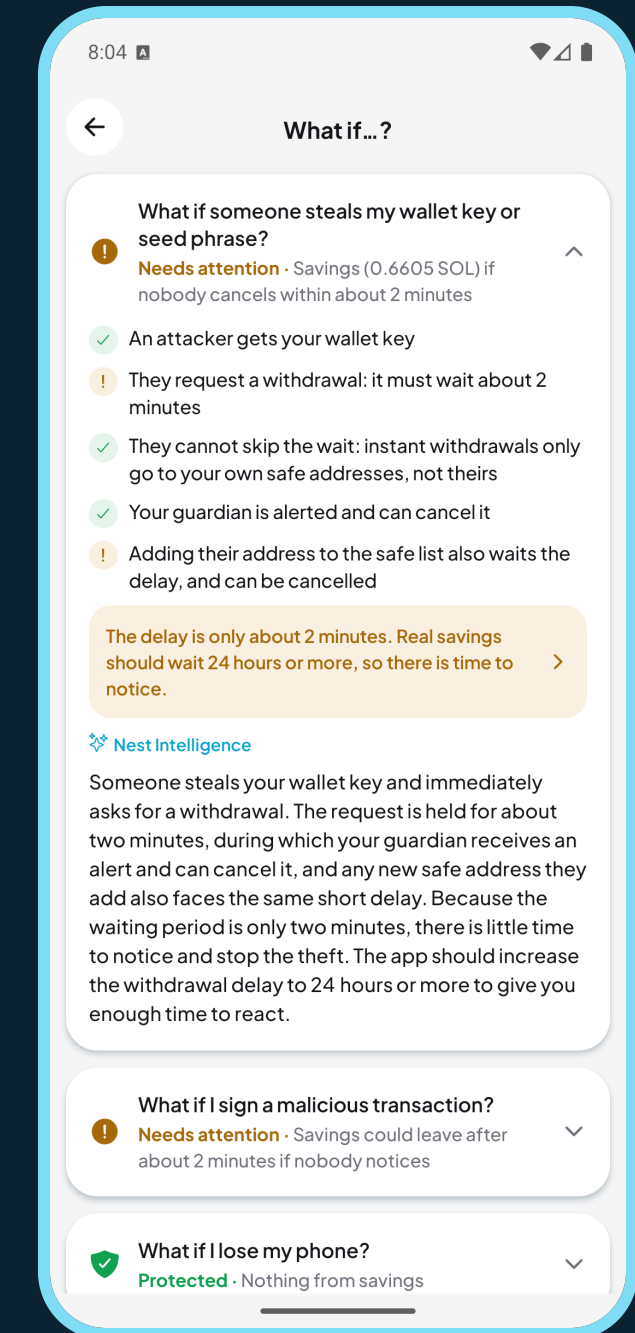
AI explains, scores and simulates. The program decides.

- **Risk scoring:** every withdrawal scored from on-chain signals; guardians get a briefing, and the alert says "High risk".
- **Transaction explainer:** what you are about to sign, in plain words, before the wallet opens.
- **What-if simulator:** seven attacks played against your real setup, with fixes and free-form questions.

Scores and outcomes are computed in code; Groq only words them.
No keys, addresses or location are sent.



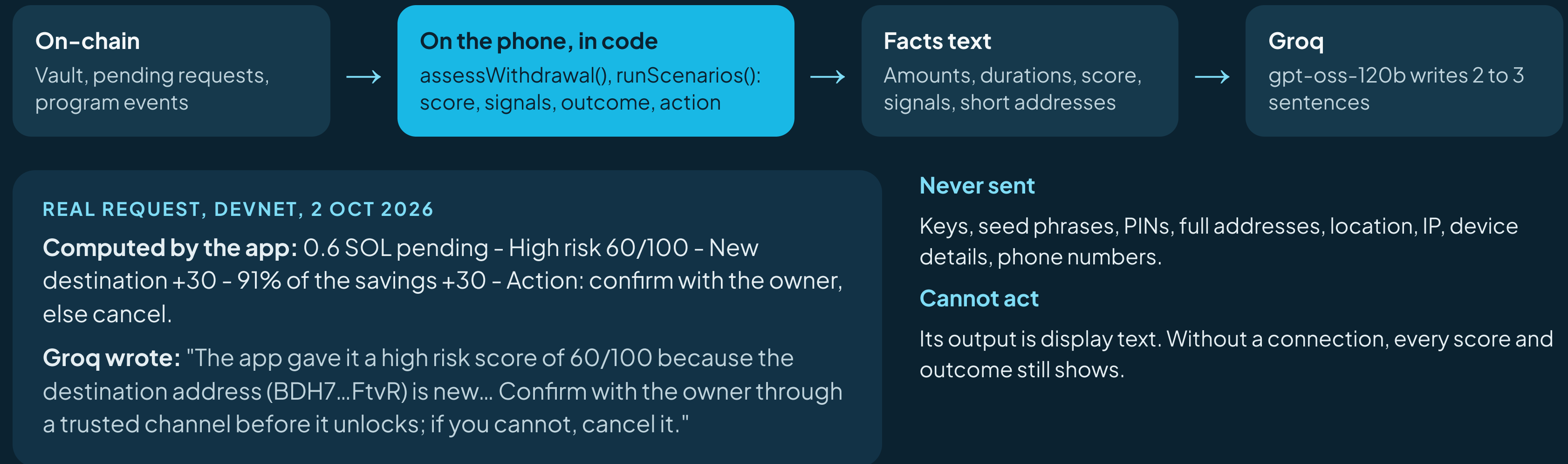
Guardian briefing



What-if simulator

AI DATA FLOW

The app decides. Groq only words what it decided.

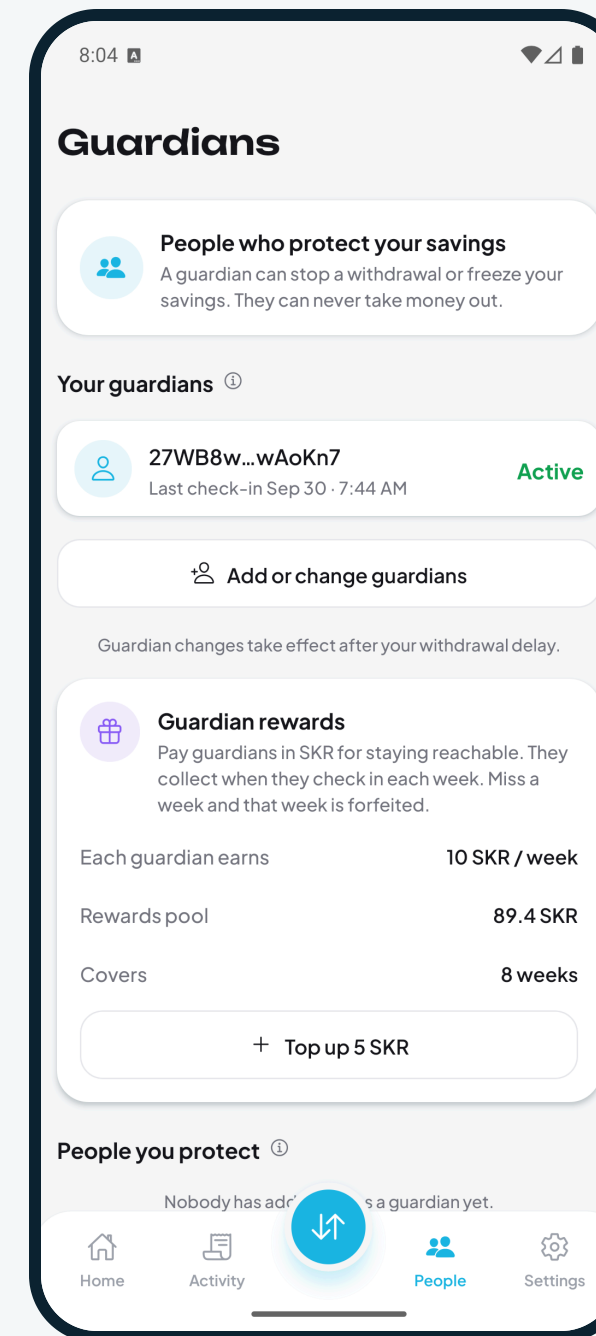


SKR

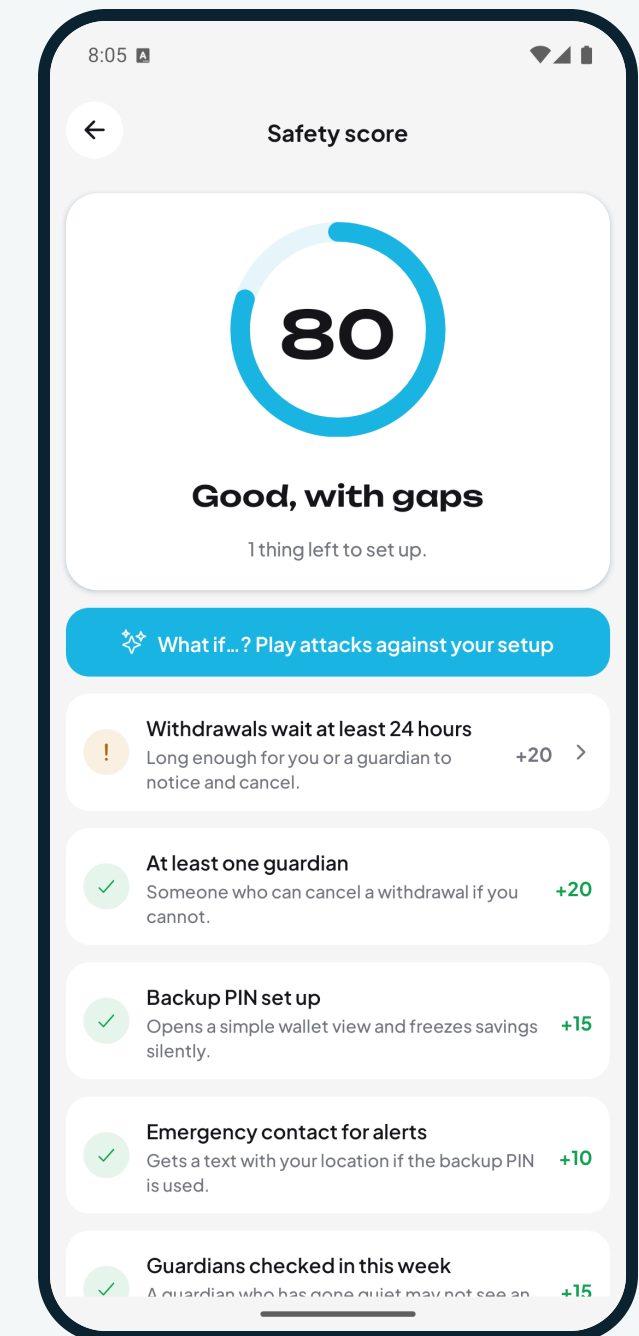
Guardians earn SKR for staying reachable.

- The owner funds a rewards pool in SKR.
- Guardians collect when they check in each week.
- Each claim covers at most 8 days, so a missed week is forfeited.
- The safety score shows gaps, from delay length to funded rewards.

Tested against SKR's real mainnet mint account. Mainnet is a config change, not a code change.



Rewards pool



Safety score

A full reward cycle, traced on devnet.

1

Pool funded

setup_stipend + fund_stipend:
5 tokens into the pool's own
account, at 1 token a week.

tx 5ZVcKv...BgHz1L

2

First check-in

claim_stipend pays 1 token (a
first claim counts as one week)
and records the heartbeat.

tx q2cKdV...WX4xoa

3

152 s later

Pays 251 base units: $1,000,000 \times 152 \div 604,800 = 251$, exactly
the formula.

tx 4q8Bf2...qWP5TJ

4

Heartbeat

guardian_last_seen
1790922408 → 1790922560,
read back from the vault.

same transaction

Rules the program enforces

$\text{paid} = \text{rate} \times \min(\text{time since last claim}, 8 \text{ days}) \div 1 \text{ week}$, never more
than the pool holds. Only current guardians can claim. The pool is
separate from the savings.

Mainnet switch

Devnet uses Circle's devnet USDC as the stand-in (same Token
program, 6 decimals). Mainnet sets the mint to SKR
(SKRbvo6Gf7GondiT3BbTfuRDPqLWei4j2Qy2NPGZhW3): a
config value, tested against SKR's real mint account.

BUILT FOR SEEKER

Built for the phone, not ported to it.

Seed Vault + MWA

The owner key never leaves Seed Vault. Every action is approved in the wallet sheet.

Sentinel key

A tighten-only key in Android Keystore signs the silent freeze.

Silent SMS

A small Kotlin module sends the alert with nothing on screen. Android only.

Real data only

The backup-PIN view shows the real wallet, so an explorer check matches.

Was this you?

An alert for any withdrawal this phone didn't request.

Native app

Expo and React Native with @solana/kit and a typed client generated from the IDL.

PROOF

Working on devnet today.

22 / 22

program tests pass on the deployed binary, including an attacker holding the owner's real key, and guardian rewards on SKR's real mainnet mint.

65 devnet transactions, decoded and linked in [docs/devnet-evidence.md](#):

- Vaults created, deposits, withdrawals after the delay
- Withdrawals cancelled by owner and guardian
- Backup-PIN freezes signed by the phone's own key
- Settings change waited out, then applied
- Rewards pool funded and collected on check-in

JUDGING CHECKLIST

Where to find each thing.

Criterion	Evidence	Where
Runs on Android, with MWA	APK on GitHub; every signature through Mobile Wallet Adapter; sentinel key in Android Keystore	GitHub release; use-vault.ts; sentinel.ts
Stickiness	Weekly guardian check-ins paid in SKR; risk alerts; safety score and What-if simulator	skr.md; simulator.ts
User experience	Backup PIN decoy, one-tap cancel, plain-language AI, faucet buttons and demo timers for judges	README: For judges
Innovation	Duress PIN + on-chain time lock + a key that can only tighten, so a coerced unlock gives nothing	threat-model.md
Demo	2:50 video of the flows on devnet	youtube.com/shorts/_GFJuXn5x9o
SKR integration	Rewards pool enforced on-chain; full cycle traced; tested on SKR's mainnet mint	skr-trace.md
Technical evidence	22 of 22 tests; deployed binary matches the repo; IDL on-chain; 65 decoded transactions	EVIDENCE.md
AI, honestly scoped	Scores computed in code; Groq words them; exact payload published	ai.md

What works today, and what comes next.

Implemented, on devnet

- Anchor vault: delays, safe list, freezes, delayed settings
- Android app with Mobile Wallet Adapter
- Backup PIN: silent freeze by the Keystore key, SMS, freeze report
- Guardians: discovery, risk-scored alerts, one-tap cancel
- SKR rewards, tested on SKR's real mainnet mint
- Nest Intelligence: risk scoring, briefings, What-if simulator
- 22 tests on the deployed binary, 65 devnet transactions

Planned

- Professional audit, then mainnet with real SKR
- Upgrade authority made immutable or a multisig
- Guardian alerts delivered in the background (push)
- AI through a server-side proxy, so no key ships in the app
- Owner can close a rewards pool and reclaim unspent SKR
- A freeze cooldown per role, so a hostile guardian cannot keep re-freezing
- Listing on the Solana dApp Store

TEXT VERSION – SPEAKER NOTES

1. Nest Finance

Nest Finance is a savings app for the Solana Seeker. It looks like an ordinary personal finance app, but your savings sit in an on-chain vault that nobody can rush, not even someone holding your unlocked phone.

2. A Seeker in your hand says you hold crypto.

Owning a Seeker tells people you hold crypto. If someone forces you to unlock it, or you sign one bad transaction, everything can leave in seconds, and nothing in a normal wallet slows that down.

3. Even with your real key, nobody can rush your savings.

Nest Finance makes that encounter pointless. Savings sit in a vault where every withdrawal waits. A second PIN opens an ordinary-looking wallet and quietly freezes the savings. And guardians, people you trust, can cancel a withdrawal before it lands.

4. The program enforces the rules, not the app.

An Anchor program on Solana holds the savings. Withdrawals wait a delay; the only instant path is to your own safe addresses; even settings changes wait. There are three keys: the owner in Seed Vault, guardians on their own phones, and a sentinel key in this phone's keystore that can only tighten, never move funds.

TEXT VERSION – SPEAKER NOTES

5. Two phones, one program. The chain holds the money and the rules.

The owner's phone runs the Nest Finance app. The owner's key stays in Seed Vault or their wallet and signs through Mobile Wallet Adapter. The app also holds a sentinel key in Android Keystore that can only freeze and cancel, so the backup PIN can freeze the vault with no wallet prompt and text an emergency contact. The savings and every rule live in the nest_vault program on Solana. Guardians run the same app on their own phones: it watches the vault, raises a risk-scored alert, lets them cancel or freeze with one tap, and pays a weekly SKR check-in. Groq sits off to the side: it receives facts text and returns wording, nothing else.

6. 5 account types, 17 instructions, each with fixed signers.

The program has five account types, all derived from the owner's address, and seventeen instructions, each with a fixed set of signers. The owner alone can request and propose; the owner and a guardian together can release early or lift a freeze; any role can cancel or freeze; guardians check in and claim; anyone can deposit, or run a withdrawal whose delay has passed. Only three instructions can move savings out, and a guardian or the sentinel cannot use any of them alone.

7. The delay cannot be skipped, even with the owner's real key.

This is why the delay holds even against someone with the owner's real key. The program sets the unlock time, the payout is pinned to the stored destination, instant payouts only reach addresses already on the safe list, and any change to the safe list, the delay or the guardians waits the current delay in plain view of the guardians. The table shows what each compromised key can and cannot do. One honest gap: a hostile guardian can keep re-freezing the vault. That holds savings up but never takes them, and a freeze cooldown is planned.

8. Two PINs. One opens an ordinary wallet.

If you're forced to open the app, enter your backup PIN. It opens what looks like a normal wallet with only your spending money, all real data. Behind the scenes the phone's own key freezes your savings with no wallet prompt, and your emergency contact gets a text with your location.

TEXT VERSION – SPEAKER NOTES

9. Someone you trust can stop it in time.

A guardian opens the same app with their own wallet, and Nest finds every vault that lists them. When a withdrawal is requested, maybe by a drainer, maybe under pressure, the guardian gets an alert and can cancel it before the delay runs out.

10. AI explains, scores and simulates. The program decides.

Nest Intelligence makes the security understandable. Every withdrawal is scored for risk from on-chain signals, like a destination never used before or most of the savings leaving at once, and guardians get a plain-language briefing, with the risk level right in the alert. Before signing, the owner sees what the transaction does. And the What-if simulator plays seven attacks against the owner's real setup and says what to fix. The principle: the app computes the scores and outcomes, AI only words them, and the Solana program decides what is allowed.

11. The app decides. Groq only words what it decided.

This is exactly what goes where. The app reads the vault and its pending requests from the chain, and plain code computes the risk score, the signals and the recommended action. Only a short facts text goes to Groq, which writes two or three sentences. Here is a real request from devnet: the app scored a 0.6 SOL withdrawal High, 60, because the destination was new and it was 91 percent of the savings; Groq only put that into words. Keys, PINs, full addresses and location are never sent, and the model can do nothing but show text.

12. Guardians earn SKR for staying reachable.

A guardian is only useful if they're reachable. Owners fund a pool in SKR, and guardians collect it when they check in each week. The program caps every claim at 8 days, so skipping a week forfeits that week. The safety score keeps owners honest about gaps. A program test loads SKR's actual mainnet mint account and runs the whole cycle, so moving to mainnet is a configuration change.

TEXT VERSION – SPEAKER NOTES

13. A full reward cycle, traced on devnet.

Here is one complete rewards cycle on devnet, read back from the chain after each step. The owner funds a pool. The guardian's first check-in pays one week. A second check-in 152 seconds later pays 251 base units, exactly what the formula gives, and both update the guardian's heartbeat on the vault. The 8-day cap is checked by a program test that skips ahead 30 days. On mainnet the same program runs with the SKR mint.

14. Built for the phone, not ported to it.

This only works on a phone. The owner key stays in Seed Vault, a tighten-only sentinel key lives in the Android keystore, and a small native module sends the alert text silently, something iOS apps can't do.

15. Working on devnet today.

Everything you've seen runs on devnet today. The program has 22 tests, run against the exact binary deployed on devnet: one has an attacker holding the owner's real key try every route to the money, and one runs guardian rewards on SKR's real mainnet mint account. On devnet there are 65 decoded transactions on the demo vaults, each linked in the repository's evidence page.

16. Where to find each thing.

For judges, this maps each criterion to its evidence: the APK and the Mobile Wallet Adapter code, the weekly SKR check-ins that bring guardians back, the backup-PIN experience, the on-chain design behind it, the demo video, the SKR trace, the tests and deployment proof, and the AI payload. Everything is in the repository, starting from EVIDENCE.md.

TEXT VERSION – SPEAKER NOTES

17. What works today, and what comes next.

Here is exactly what is built and what is not. Built and running on devnet today: the vault with all its rules, the Android app, the backup PIN with its silent freeze and alert, guardians with risk-scored alerts, SKR rewards tested against the real SKR mint, and Nest Intelligence. Next: an audit, mainnet with real SKR, a locked upgrade authority, background push alerts for guardians, moving the AI key behind a server, and a dApp Store listing. Nest Finance: savings that can't be rushed.

Nest Finance

Nest Finance lock screen on a phone

SCREEN TEXT (OCR, MACHINE-READ)

8:02
Nest Finance
Enter your PIN
Unlock to see your accounts
1
4
7
0000

2
5
8
0
3
6
9

Two PINs. One opens an ordinary wallet.

Normal view: protected balance of 0.66 SOL with the withdrawal delay, spending and protected savings accounts

SCREEN TEXT (OCR, MACHINE-READ)

8:00	\$80.38
Hi there	0.661 SOL
HGqR...bHR3	80
Protected Balance	Savings protection
\$80.38	Active
0.6605 SOL Protected • 2m	Withdrawal delay
+ Deposit	Guardians
Accounts	Safe addresses
6	Safety score
Spending ○	2 minutes
\$610.84	1 connected
5.019 SOL	Home
Protection	Activity
Withdraw	○○
2m protection	People
View All >	Settings
Protected Savings	

Backup PIN view: an ordinary wallet showing only the 5.02 SOL spending wallet and its real transactions

SCREEN TEXT (OCR, MACHINE-READ)

8:16	Yesterday 8:27
Hi there	Receive
HGqR...bHR3	View All >
Total Balance	USDC
\$608.49	\$0.00
5.0179 SOL	0.00USDC
'Tx Send	View All >
Accounts	-0.001 SOL
6	Wallet
Main Wallet	-0.001 SOL
\$608.49	Wallet
5.018 SOL	-0.001 SOL
Recent Transactions	Home
Sent	Activity
Today • 8:09 AM	○○
Sent	People
Today • 7:30 AM	Wallet
Sent	Settings

Someone you trust can stop it in time.

Guardian alert: Withdrawal requested, High risk, 0.6 SOL from savings you protect

SCREEN TEXT (OCR, MACHINE-READ)

8:09 Fri, Oct 2
Internet
Flashlight
* Bluetooth
e Modes

Withdrawal requested • High risk • now n
0.6 SOL from savings you protect for
HGq...
Manage
Clear all

Guardian view: 0.6 SOL leaving in 1:35 with a Cancel button, High risk 60 and the AI briefing

SCREEN TEXT (OCR, MACHINE-READ)

8:09
Protected savings
HGqR25...jobHR3's savings
\$80,26
0.6605 SOL • Protected
Withdrawal requests
O
0.6 SOL to 2By2...vylc
Leaves in
@ Cancel this withdrawal
A High risk 60
Nest Intelligence
New destination. 2By2...vylc has not
been used by
these savings before today.
Most of the savings. 91% of what is in the
vault.
A withdrawal of 0.6 SOL is scheduled to
unlock

in less than an hour, and you can cancel it
any
time before it unlocks, though
cancellation does
not move any money. The app gave it a
high risk
score of 60/100 because the destination
address
is new and the amount represents 91 % of
the
vault's balance. Confirm with the owner
through
a trusted channel before it unlocks; if you
cannot,
cancel it.
Score, signals and advice computed by
the app; worded by
AI. The program decides what is allowed.
Protect

AI explains, scores and simulates. The program decides.

Guardian briefing: High risk 60, signals New destination and Most of the savings (91%), and an AI briefing advising to confirm with the owner or cancel

SCREEN TEXT (OCR, MACHINE-READ)

8:10	the
Protected savings	vault's balance. Confirm with the owner
A High risk • 60	through
Nest Intelligence	a trusted channel before it unlocks; if you
New destination. 2By2...vylc has not	cannot,
been used by	cancel it.
these savings before today.	Score, signals and advice computed by
Most of the savings. 91% of what is in the	the app; worded by
vault.	AI. The program decides what is allowed.
A withdrawal of 0.6 SOL is scheduled to	Protect
unlock	Withdrawal delay
in less than an hour, and you can cancel it	Your last check-in
any	Rewards
time before it unlocks, though	2 minutes
cancellation does	Sep 30 • 7:44 AM
not move any money. The app gave it a	10 SKR/week
high risk	V/ Checkin & collect 2.88 SKR
score of 60/100 because the destination	* Freeze their savings
address	Name
is new and the amount represents 91 % of	Who is this? (only on your phone)

What-if simulator: someone steals my wallet key, needs attention, with each step, the fix to lengthen the delay, and an AI walkthrough

SCREEN TEXT (OCR, MACHINE-READ)

8:04	Nest Intelligence
What if... ?	Someone steals your wallet key and
What if someone steals my wallet key or	immediately
seed phrase?	asks for a withdrawal. The request is held
O	for about
Needs attention • Savings (0.6605 SOL)	two minutes, during which your guardian
if	receives an
nobody cancels within about 2 minutes	alert and can cancel it, and any new safe
An attacker gets your wallet key	address they
They request a withdrawal: it must wait	add also faces the same short delay.
about 2	Because the
minutes	waiting period is only two minutes, there
They cannot skip the wait: instant	is little time
withdrawals only	to notice and stop the theft. The app
go to your own safe addresses, not theirs	should increase
Your guardian is alerted and can cancel it	the withdrawal delay to 24 hours or more
Adding their address to the safe list also	to give you
waits the	enough time to react.
delay, and can be cancelled	What if I sign a malicious transaction?
The delay is only about 2 minutes. Real	Needs attention • Savings could leave
savings	after
should wait 24 hours or more, so there is	about 2 minutes if nobody notices
time to	What if I lose my phone?
notice.	Protected Nothing from savings

Guardians earn SKR for staying reachable.

Guardian rewards card: 10 SKR per week

SCREEN TEXT (OCR, MACHINE-READ)

8:04	collect when they check in each week.
Guardians	Miss a
People who protect your savings	week and that week is forfeited.
A guardian can stop a withdrawal or	Each guardian earns
freeze your	Rewards pool
savings. They can never take money out.	Covers
Your guardians	+ Topup5SKR
o	People you protect
27WB8w...wAoKn7	10 SKR/week
Active	89.4 SKR
Last check-in Sep 30 7:44 AM	8 weeks
+0 Add or change guardians	Nobody has add
Guardian changes take effect after your	s a guardian yet.
withdrawal delay.	Home
Guardian rewards	Activity
Pay guardians in SKR for staying	People
reachable. They	Settings

Safety score of 80

SCREEN TEXT (OCR, MACHINE-READ)

8:05	cannot.
Safety score	Backup PIN set up
80	Opens a simple wallet view and freezes
Good, with gaps	savings
1 thing left to set up.	silently.
What if... ? Play attacks against your	Emergency contact for alerts
setup	Gets a text with your location if the
Withdrawals wait at least 24 hours	backup PIN
Long enough for you or a guardian to	is used.
notice and cancel.	Guardians checked in this week
At least one guardian	A •nrAinn hoe
+20	+20
Someone who can cancel a withdrawal if	+15
you	+10